EasyMeals

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1. Team info

Roles

- Dean Leon - Full-stack developer
- Ayowade Owojori - Backend lead
- Dylan Peniuk - Project leader
- Noam Yaffe - Frontend lead

GitHub Repository

- https://github.com/yaffenator/EasyMeals

Communication Channels

- Microsoft Teams

2. Product description

Abstract

Meal prepping is hard. For any individual receiving food benefits or other financial aid, the idea of properly allocating a portion of their monthly income to groceries as effectively and efficiently as possible may seem like a daunting task. Introducing: EasyMeals – Automate your meal-prepping goals with a personalized AI nutrition expert. This website will allow you to enter the amount of money you allocate to food and groceries each month and select your physical goals (such as losing weight or building muscle) and their extent. Using this information, EasyMeals will create a month-long meal-prepping plan, showing you what groceries to buy, how much of each grocery item you will need, and some fun meals that you can prepare to satisfy your hunger.

Goal

The goal of this project is to make a website that allows people to skip the trouble of budgeting for food and the stress of finding recipes to prepare, by having it all in one place. It will allow people to enter their monthly food budget and automatically give a list of groceries to buy and recipes of what to cook week by week. It will also ask for food allergies and restrictions to let people be stress-free about what they are eating. Overall, this project would reduce the stress of wondering what you are able to afford and what you are able to make, by laying it all out month by month.

Current practice

Today, this is similarly found in meal plans like Hello Fresh, where you pay a monthly subscription, and they deliver ingredients to your house. The problem with this is that it is expensive, and it's impossible to know where that food is really coming from. Also, for lower-income families who rely on government aid programs like EBT/food stamps, they are unable to use that money for those programs.

Novelty

What makes our approach different is the introduction of an AI assistant that will webscrape products from local grocery stores to create specialized meal plans that are applicable to user requests. This approach will factor in the price of the ingredients, which is absent from existing services, and the ability to cook the meals yourself, where services like Factor_ ship the entire meal.

Effects

Everyone should care about their expenses with food. However, this is targeted more towards people with lower incomes, or multiple jobs, who worry about having enough money for food or spending the time to research what to make. This would make a difference because people would have easy access to find meals that work for their budget and schedule.

Technical approach

To build this meal-prepping website, we will use the React library combined with a Next.js framework. We chose this tech-stack as it will allow us to leverage features like automatic code splitting, server-side rendering, and more. We plan to use TypeScript (+ TailwindCSS as a replacement for HTML/CSS) for our frontend components and Python for our backend functions. We will send and receive information from our fine-tuned AI model, which will most likely be implemented via a Gemini API handler, and MongoDB (or Firebase, Supabase, etc.) as our database to store individual user information and preferences. If time permits, we also plan to implement a user authentication system using both basic email/password authentication and more advanced OAuth (such as Google and other OAuth providers).

Risks

The biggest risk of this project is making sure that we find the prices of food accurately, so we don't put people over budget. Another risk is making sure we fully exclude all items that are flagged as food restrictions as not to be liable for an allergic reaction or other response to food the user shouldn’t be eating. Another risk is that our program will collect sensitive user data (weight, lifestyle, ect.), so our program should be secure enough to ensure that this data is kept private.

Additionally, add the following:

- 4+ major features you will implement.
    - Weekly grocery list that stays in budget of the user
    - A full month of recipes for meals the user can cook at home
    - Meals targeted towards fitness goals
    - Allowing the user to add food restrictions to prevent allergic reactions
- 2+ stretch goals you hope to implement.
    - Allow the user to add family members to their meal plan (prep for a family of 4)
    - Allow certain selection of grocery stores to purchase from nearby

The major features should constitute a minimal viable product (MVP).